

THE SYNTRACT VISION

QCDS x DNA

THE INFERENCE FABRIC

Rotational Nulling / Positional Rotation / Oracle Rotation / Recursive Funnel /
Syntract Bind

SERIOUS ARCHITECTURE. STAGE-SCALE ENERGY.

Patrik Sundblom

THE PROBLEM

**DNA IS NOT A FILE.
IT IS A LIVING COMPOSITIONAL
SYSTEM.**

WHY QCDS

THE SEQUENCE IS THE EASY PART.

The hard problem is simultaneous interaction across variants, regulation, expression, proteins, cells, environment and time.

VARIANT x VARIANT

REGULATION x EXPRESSION

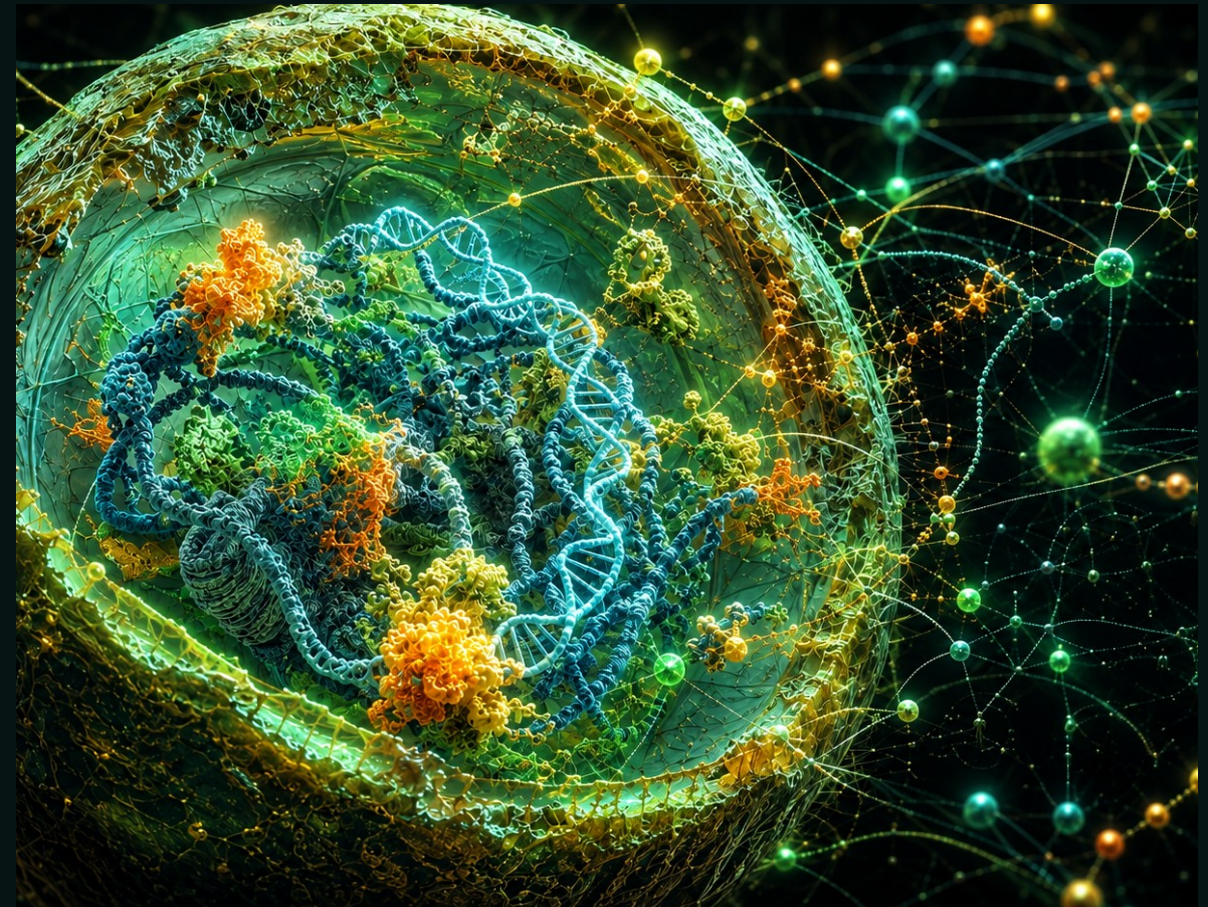
PROTEIN x PATHWAY

CELL x TISSUE

ENVIRONMENT x TIME

PHENOTYPE x INTERVENTION

QCDS is aimed at the structure of the possibility space - not merely the storage size of the genome.





THE SHIFT

**DON'T COMPRESS THE WORLD FIRST.
DISTRIBUTE THE INFERENCE.**

ARCHITECTURE

FIVE WORDS. NO AMBIGUITY.

The hierarchy has to be explicit before the machine makes sense.

DIMENSION

One logical variable: b0, b1, ...

CHANNEL

One QCDS inference view of a B-dimensional bundle.

BUNDLE

B parallel channels; each nulls a different dimension.

LAYER

Many bundles executing in parallel.

FUNNEL

Layer returns become higher-order dimensions for the next layer.

ORACLE TOPOLOGY

SAME ORACLES. BEFORE EVERY CHANNEL.

The oracle definition is replicated at each channel boundary. Null-aware evaluation changes only which dimension participates.



Same constraints. Same bundle. Different excluded dimension in each channel.

ROTATIONAL DIMENSION NULLING

ONE BUNDLE. B CHANNELS. B DIFFERENT NULLS.

Rotation means the exclusion advances across dimension identity - not that the same hole is shuffled through positions.

CH01	∅	b1	b2	b3	b4	b5	b6	b7
CH02	b0	∅	b2	b3	b4	b5	b6	b7
CH03	b0	b1	∅	b3	b4	b5	b6	b7
CH04	b0	b1	b2	∅	b4	b5	b6	b7
CH05	b0	b1	b2	b3	∅	b5	b6	b7
CH06	b0	b1	b2	b3	b4	∅	b6	b7
CH07	b0	b1	b2	b3	b4	b5	∅	b7
CH08	b0	b1	b2	b3	b4	b5	b6	∅

8-BIT EXAMPLE

8

dimensions in bundle

8

parallel channels

8

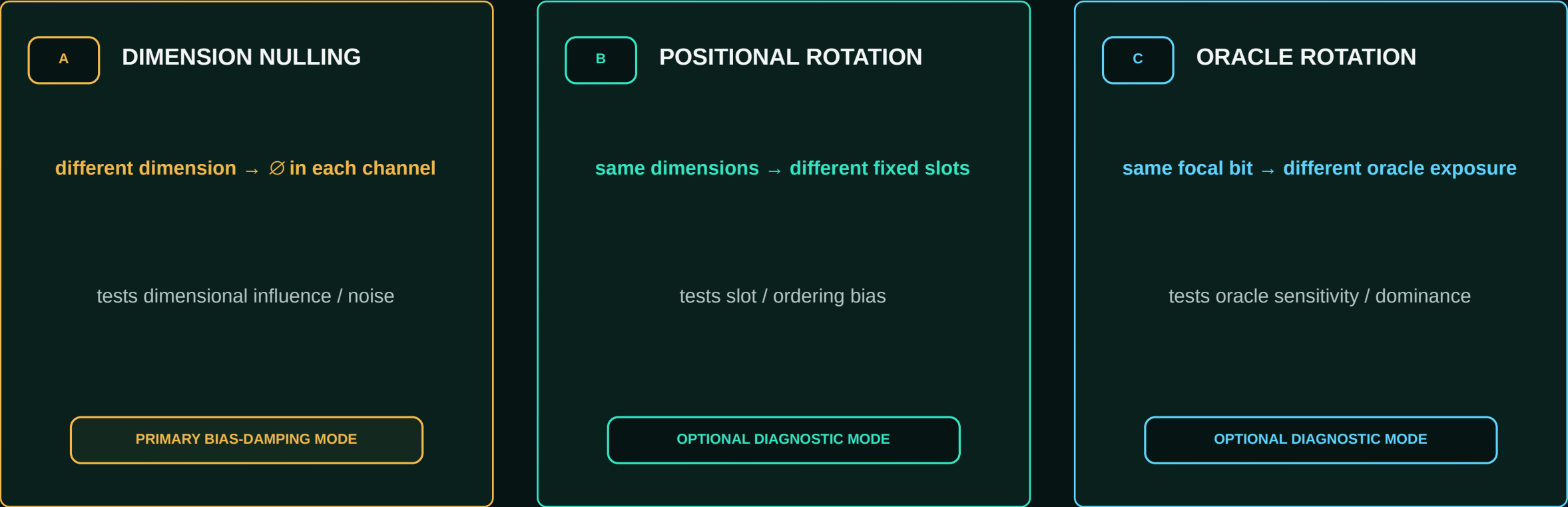
unique null views

CH01 nulls b0
CH02 nulls b1
...
CH08 nulls b7

ROTATION FAMILY

ROTATION IS A TOOLBOX - NOT ONE MOVE.

QCDS can rotate different axes depending on what must be tested: dimension identity, slot position, or oracle exposure.



Rotate one axis at a time - or cross axes when the experiment requires it. Do not conflate the mechanisms.

ROTATION MODE B

POSITIONAL ROTATION TESTS THE MACHINE - NOT THE DIMENSION.

Hold the same dimensions constant. Move them through fixed logical / physical slots. If the return changes, the slot is part of the bias.

R0	[b0][b1][b2][b3][b4][b5][b6][b7]
R1	[b7][b0][b1][b2][b3][b4][b5][b6]
R2	[b6][b7][b0][b1][b2][b3][b4][b5]
R3	[b5][b6][b7][b0][b1][b2][b3][b4]
...	...
R7	[b1][b2][b3][b4][b5][b6][b7][b0]

READOUT

same dimension
+ different slot
= position sensitivity

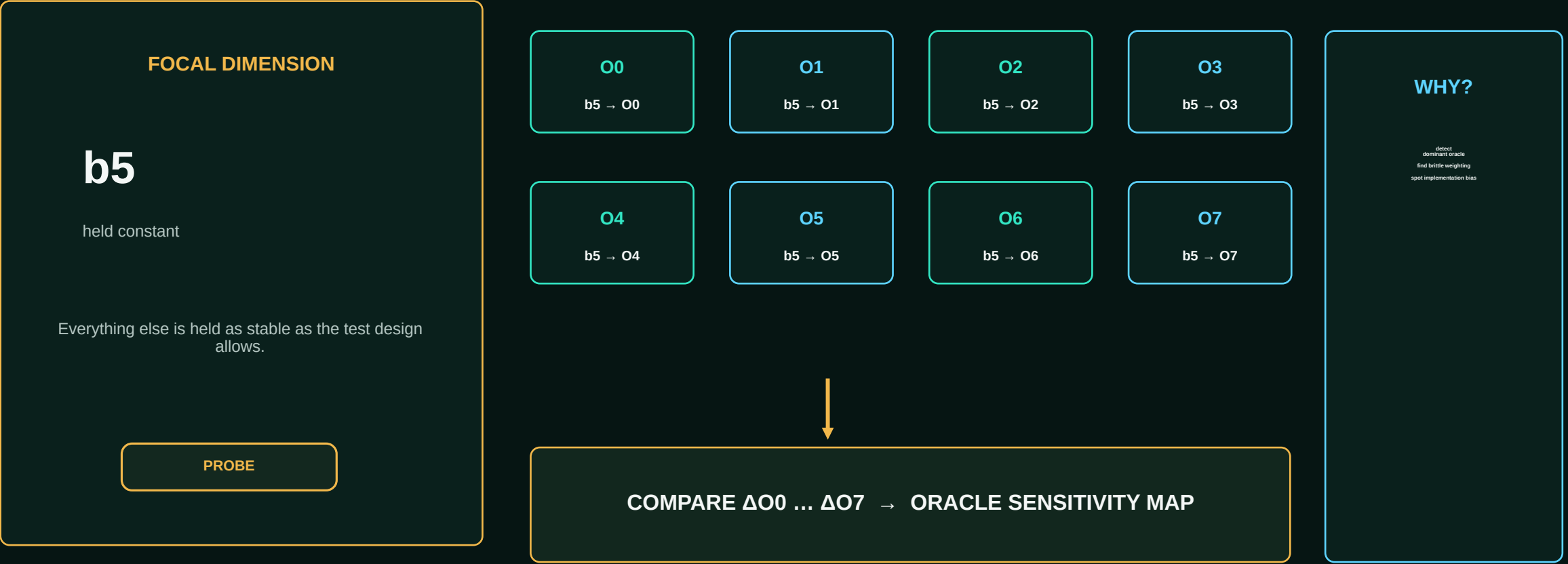
Use this to detect ordering, routing
or hardware-slot bias.

VALID ROTATION MODE

ROTATION MODE C

ROTATE THE SAME BIT THROUGH THE ORACLE STACK.

Oracle rotation is a diagnostic stress test: keep the focal dimension fixed and change which oracle / oracle position interrogates it.



Diagnostic mode - not automatically a truth run when oracle semantics are non-interchangeable.

ROTATION MATRIX

THREE AXES. THREE DIFFERENT QUESTIONS.

The same inference fabric can stress-test dimensions, positions and oracles independently - then combine them when needed.

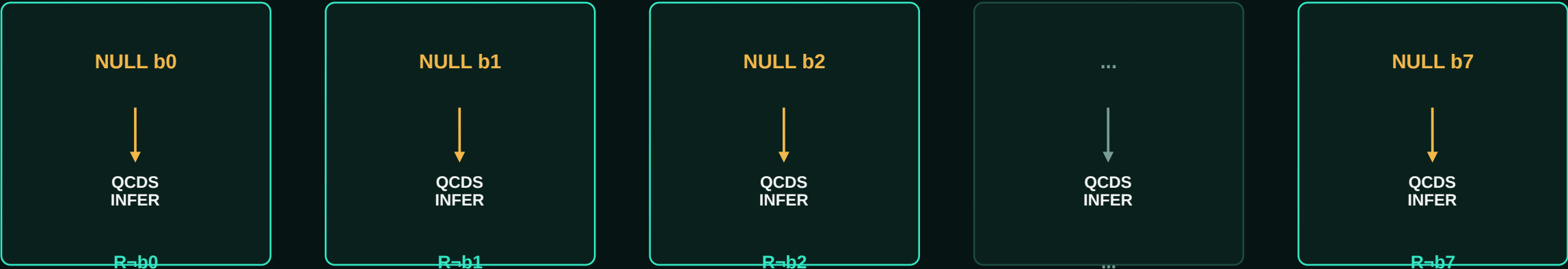
DIMENSION	which variable is absent?	NULL b0 → NULL b1 → ...	DIMENSIONAL BIAS
POSITION	where does the variable sit?	R0 → R1 → ...	SLOT / ORDER BIAS
ORACLE	which oracle interrogates it?	00 → 01 → ...	ORACLE BIAS

EXAMPLE CROSS-TEST: NULL b5 × SLOT R3 × ORACLE O2 → isolate interaction-specific bias

COUNTERFACTUAL ABSENCE

REMOVE ONE DIMENSION. ASK AGAIN.

Signal and noise are not assumed. They are tested through controlled absence under the same oracle regime.



COMPARE ALL RETURNS → DIMENSIONAL INFLUENCE MAP → DAMP BIAS / PRESERVE SIGNAL

MEASUREMENT

THE BANK PRODUCES A DIMENSIONAL INFLUENCE MAP.

Each null channel can be compared using the same return metrics. No invented answer is required.

Δ COHERENCE

Δ ENTROPY

Δ LIFT

Δ TOP-K

Δ JACCARD

Δ STABILITY

If nulling b5 increases stability and coherence across the bundle:

b5 is a candidate bias/noise dimension or harmful interaction.

If nulling b5 collapses the surviving truth basin:

b5 is likely strongly information-bearing.

The mechanism measures influence by counterfactual absence.

BUNDLE BIND

B CHANNELS IN. ONE STABILIZED RETURN OUT.

A bundle is the smallest bias-damping unit of the fabric.



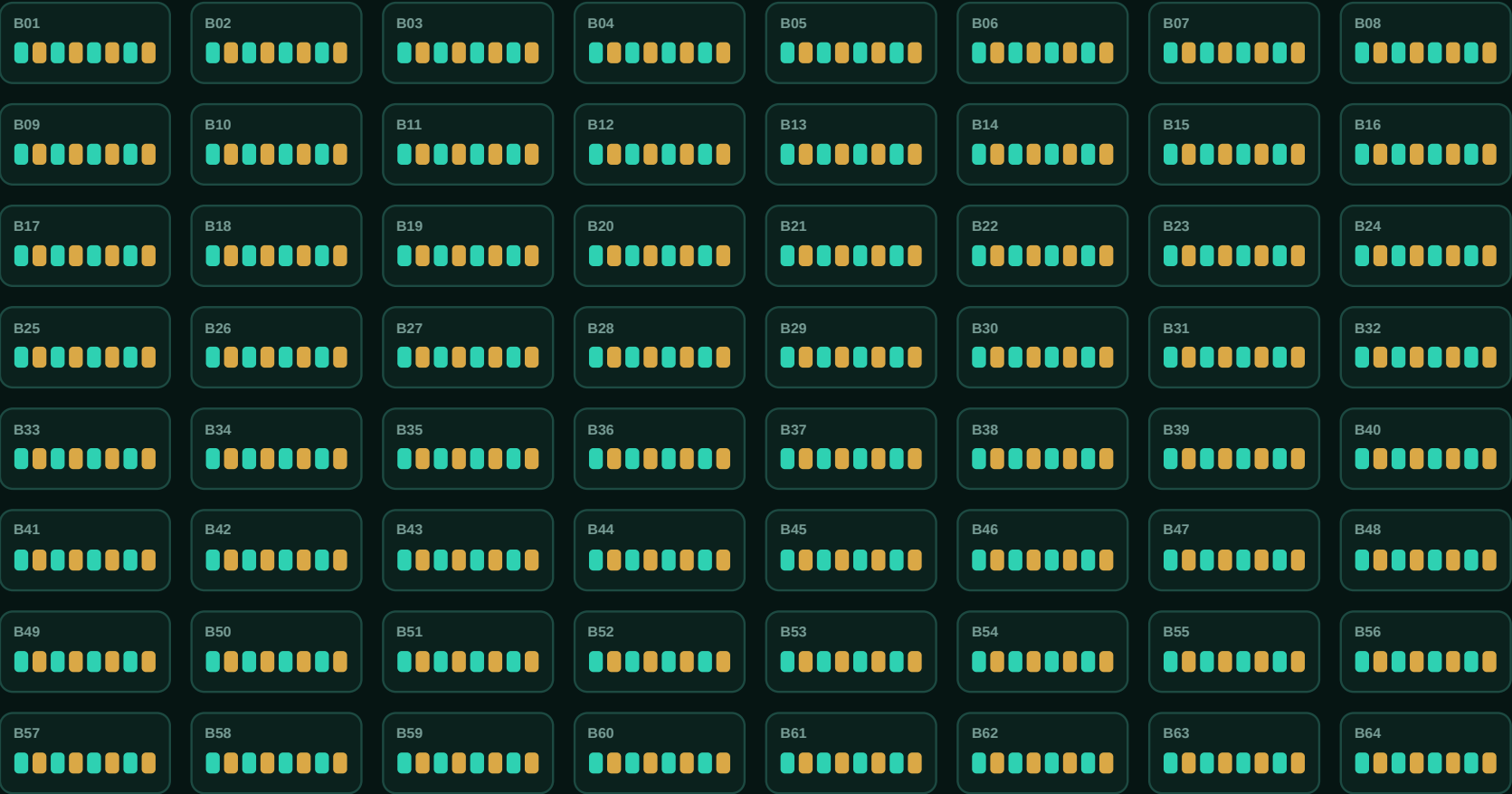
SCALE

**NOW REPEAT THE BUNDLE
HUNDREDS OF TIMES IN PARALLEL.**

512-CHANNEL INGRESS

512 CHANNELS = 64 ROTATIONAL-NULL BUNDLES AT B = 8.

Every bundle has eight unique null views. Every channel receives the same oracle definition.



64

bundles

8

channels / bundle

512

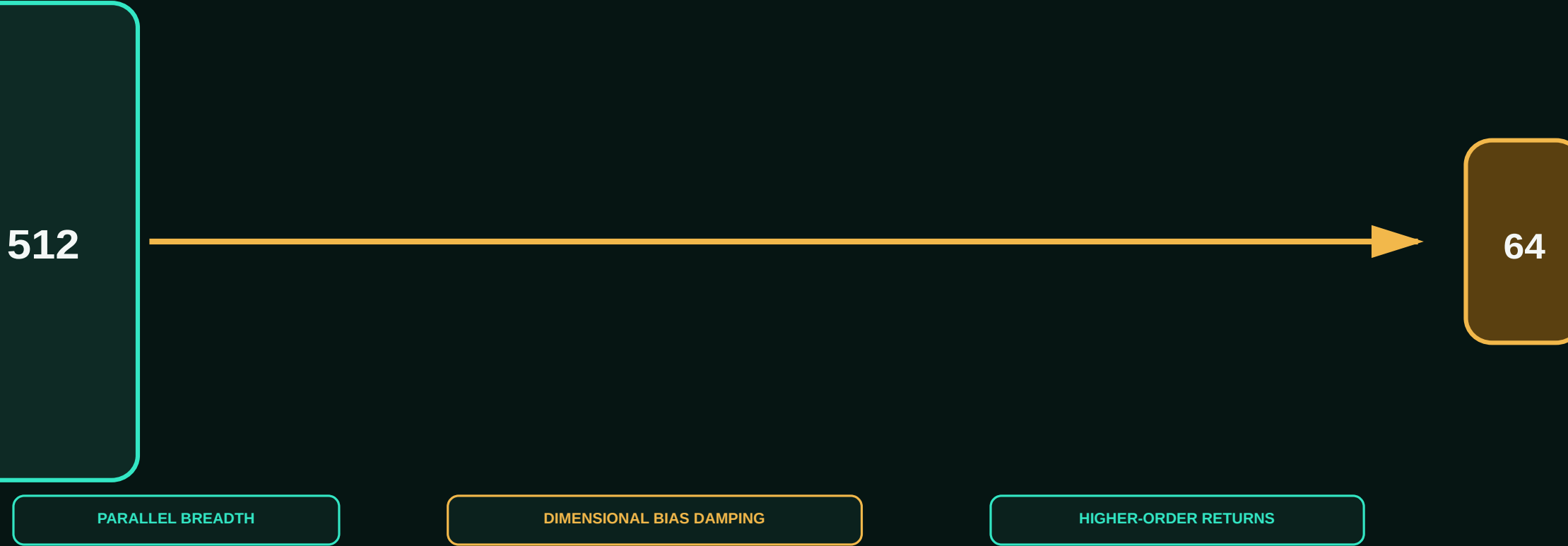
parallel channels

512 unique input dimensions
organized as 64 groups of 8.

RECURSIVE FUNNEL

THE FIRST LAYER IS 512 → 64.

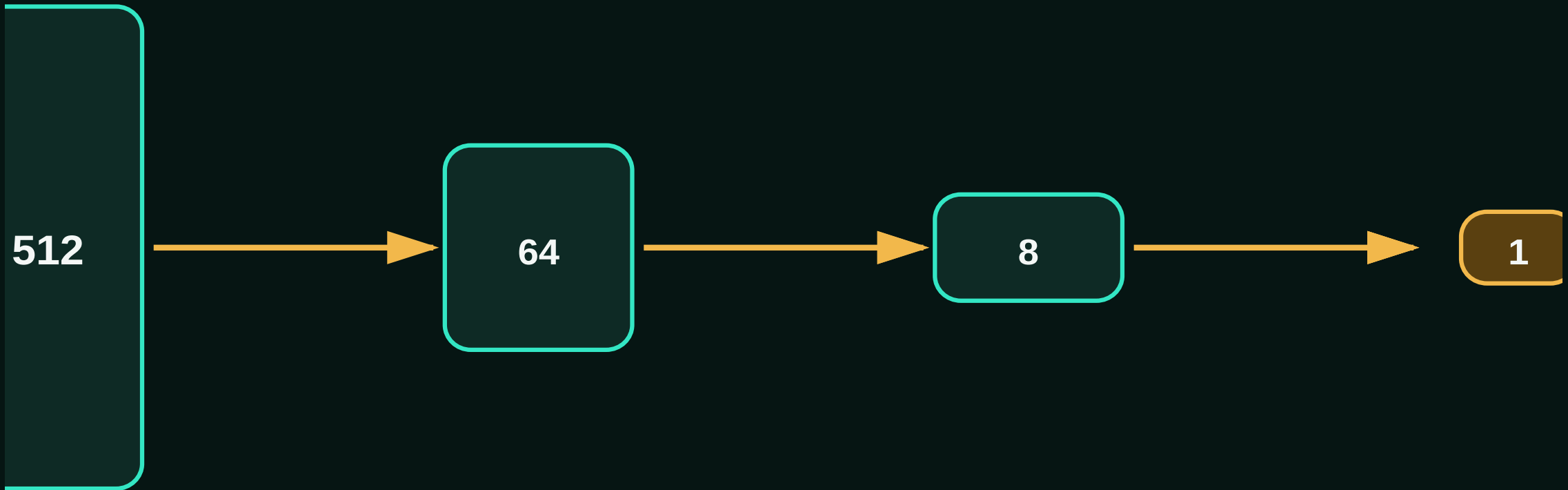
Each 8-channel bundle emits one stabilized return. That return becomes a higher-order dimension.



RECURSIVE FUNNEL

512 → 64 → 8 → 1.

Parallel inside every layer. Serial between layers. The same logic recurs at higher order.



DEEP FUNNEL

4096 → 512 → 64 → 8 → 1.

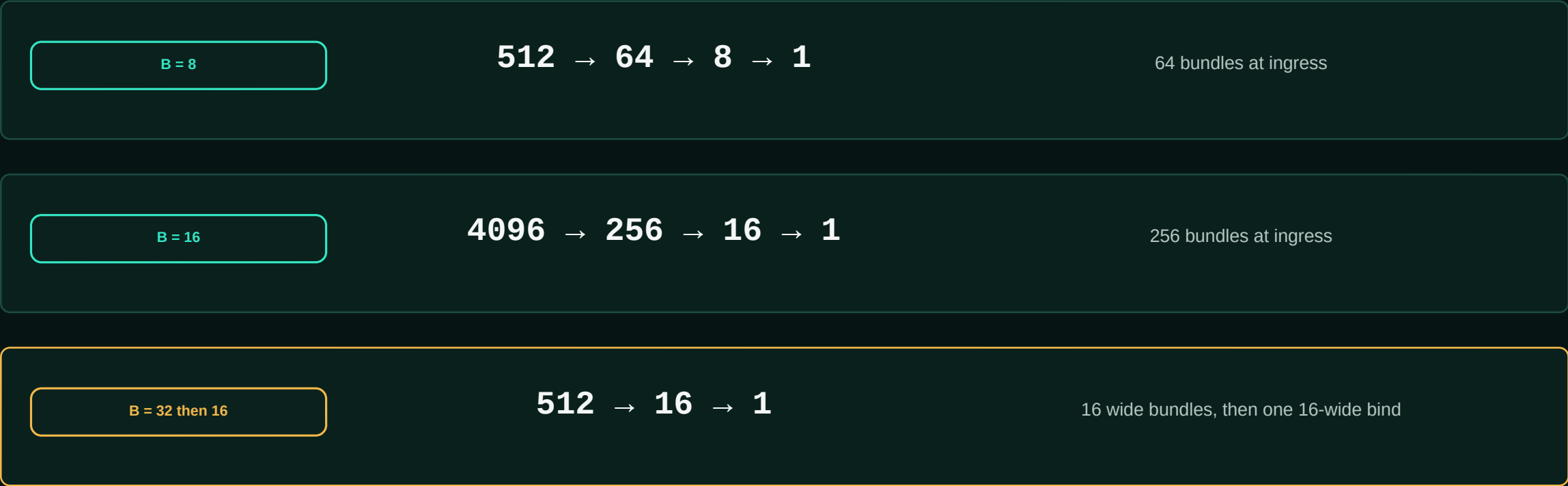
A longer compile target: huge ingress breadth, repeated rotational nulling, recursive binding.



COMPILE TARGETS

8 WAS NEVER SACRED.

Bundle width can change with substrate, latency budget and problem structure.



QCDS compiles to the machine. The machine does not define QCDS.

LOGICAL SPACE

COUNT INDEPENDENT DIMENSIONS - NOT DUPLICATED VIEWS.

The null channels are measurements of the same bundle, not new independent logical variables.

B = 8, C = 512 channels

64

bundles

512

independent binary dimensions

$2^{512} \approx 1.34 \times 10^{154}$

candidate global product space

ONE 8-DIMENSION BUNDLE

2^8

full local states

2^7

active states per null view

8 leave-one-dimension-out views
probe the same logical bundle.

Rigor matters: duplicated channel-bit positions do not magically create independent dimensions.

WIDE BUNDLES

512 CHANNELS AT $B = 32 \rightarrow 16$ STABILIZED RETURNS.

Wider local state spaces trade fewer bundles for deeper local composition. The next layer can change width.

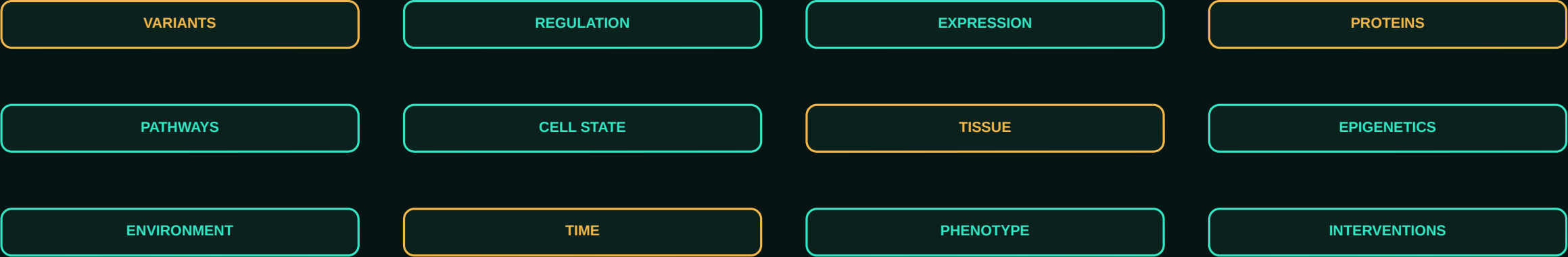


**THE FABRIC DOES NOT STORE DNA.
IT MAPS THE QUESTION INTO
DIMENSIONS.**

DNA DIMENSIONS

A GENOME QUESTION CAN OPEN HUNDREDS OF DIMENSION FAMILIES.

The input fabric can be widened without forcing everything through one early compression bottleneck.



QUESTION → DIMENSIONS → BUNDLES → NULL BANKS → STABILIZED RETURNS → RECURSIVE SYNTRACT

PATTERN IN

THE MISSION ENTERS AS CONDITIONS - NOT AS AN ANSWER.

Masks, wildcards, evidence weights and target structure define what should be tested.

PATTERN A	1	1	?	0	?	1	0	1
PATTERN B	?	1	1	?	0	?	1	1
PATTERN C	1	?	?	1	1	0	?	1

? = unconstrained

∅ = dimension absent in this channel

The same pattern/oracle regime is presented to every null channel.



ORACLE STACK

ORACLES ARE THE ACTIVE CONSTRAINT FIELD.

Every channel sees the same logic. A null-aware compiler omits only the excluded dimension from that channel's evaluation.



LOCAL QCDS KERNEL

EVERY CHANNEL RUNS A COMPLETE LOCAL INFERENCE PASS.

The fabric scales by repeating a small kernel - like a processor repeats cores.



NISQ TODAY

SMALL LOCAL CIRCUITS. HUGE FABRIC OUTSIDE THEM.

QCDS does not require one giant register to express a wide mission. The proposed path is hybrid orchestration across many small inference jobs.

1

small B per local channel

2

many channels / QPUs / simulators

3

same oracle compile across the bank

4

classical orchestration + quantum local kernel

5

recursive bind outside the local register

ARCHITECTURE $\neq$ REGISTER WIDTH



DNA STRESS TEST

CAN THE FABRIC HOLD A BIOLOGICAL WORLD TOGETHER?

RARE DISEASE

FROM ONE VARIANT TO THE WHOLE CAUSAL CONTEXT.

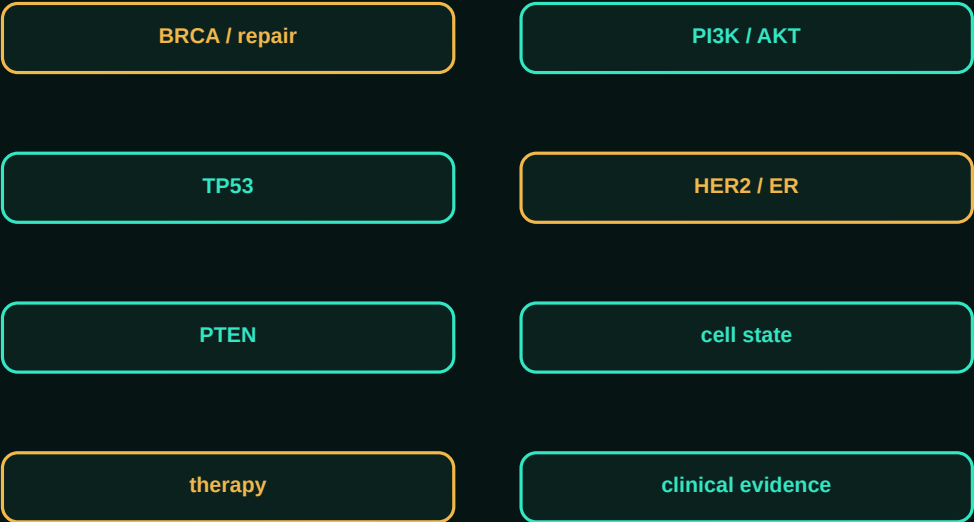
A variant is not meaningful alone. QCDS keeps its functional, cellular and evidential dimensions bound through the funnel.



CANCER

CANCER IS A MULTI-DIMENSIONAL INTERACTION PROBLEM.

Drivers, repair, signaling, expression, treatment response and contradictory evidence can remain active in parallel.



Null banks can test which dimensions actually change the surviving truth basin.



DRUG DISCOVERY

TARGET DISCOVERY BECOMES A SEARCH OVER MECHANISMS - NOT JUST CORRELATIONS.

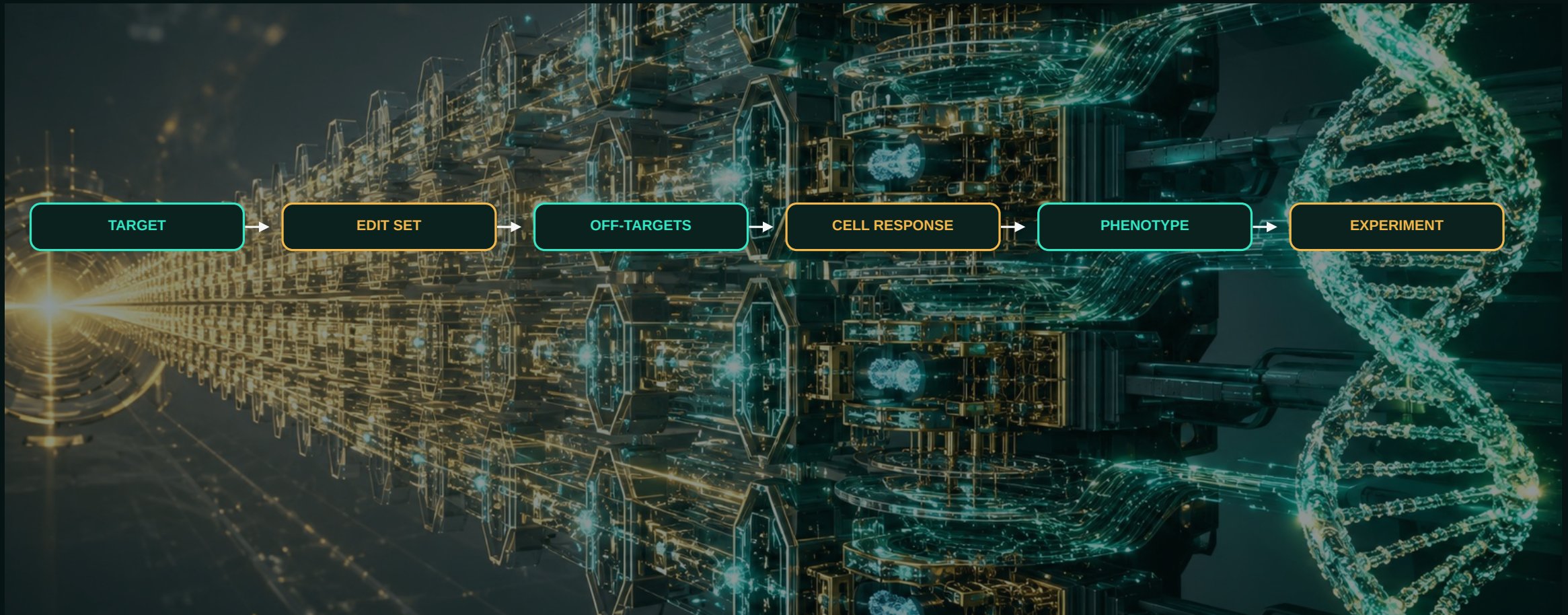
Expand from a bound disease Synttract into candidate interventions, then funnel back through evidence and experiment.



GENOME REPAIR

ASK FOR A REPAIR - THEN EXPAND THE CONSEQUENCES.

A bound target can expand into edits, pathways, off-target effects, cell responses and validation experiments.



BEYOND DNA

SELF-DRIVING IS THE SAME SHAPE OF PROBLEM.

Massive sensor breadth. Many interacting dimensions. Real-time action. The fabric cares about logical structure, not domain labels.



vision

lidar

radar

road

trajectories

weather

rules

risk

wide parallel ingress → null/bias suppression →
recursive bind → action

BEYOND DNA

TOKAMAK CONTROL IS ALSO AN INFERENCE FABRIC PROBLEM.

Plasma state is wide, dynamic and coupled. QCDS would treat control as continuous re-binding against physics and safety constraints.



magnetic field

temperature

density

current

turbulence

wall state

actuators

diagnostics

STATE → INFER → ACT → NEW STATE → RE-BIND

BIDIRECTIONAL

QCDS DOES NOT ONLY FUNNEL. IT CAN EXPAND.

Contraction asks what survives. Expansion asks what follows.

$$N \rightarrow 1$$

CONTRACT

What coherent state survives
all active constraints?

$$1 \rightarrow N$$

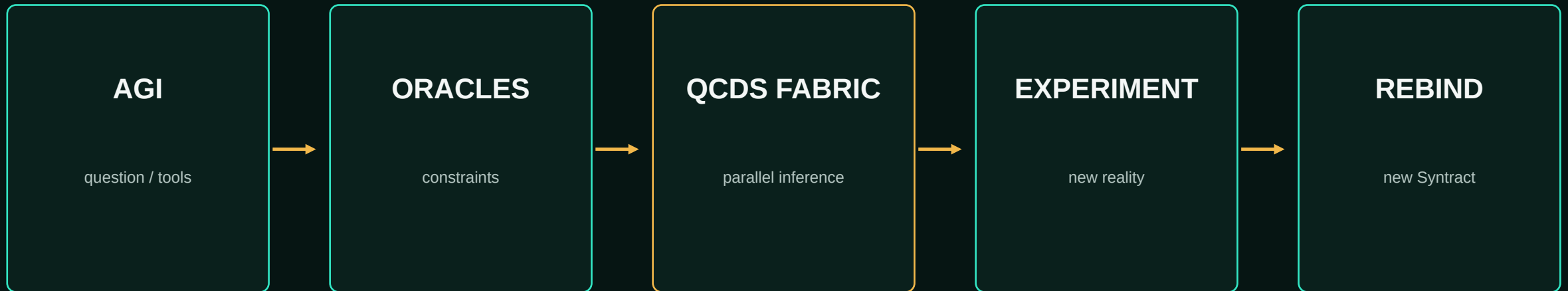
EXPAND

What mechanisms, futures,
experiments or consequences follow?

AGI → QCDS

AGI CAN FORMULATE. QCDS CAN INFER. REALITY CAN GRADE.

The proposed handoff is not another larger language model. It is an inference substrate behind autonomous scientific agency.



BETTER INFERENCE → BETTER ORACLES → BETTER QUESTIONS → NEW INFERENCE

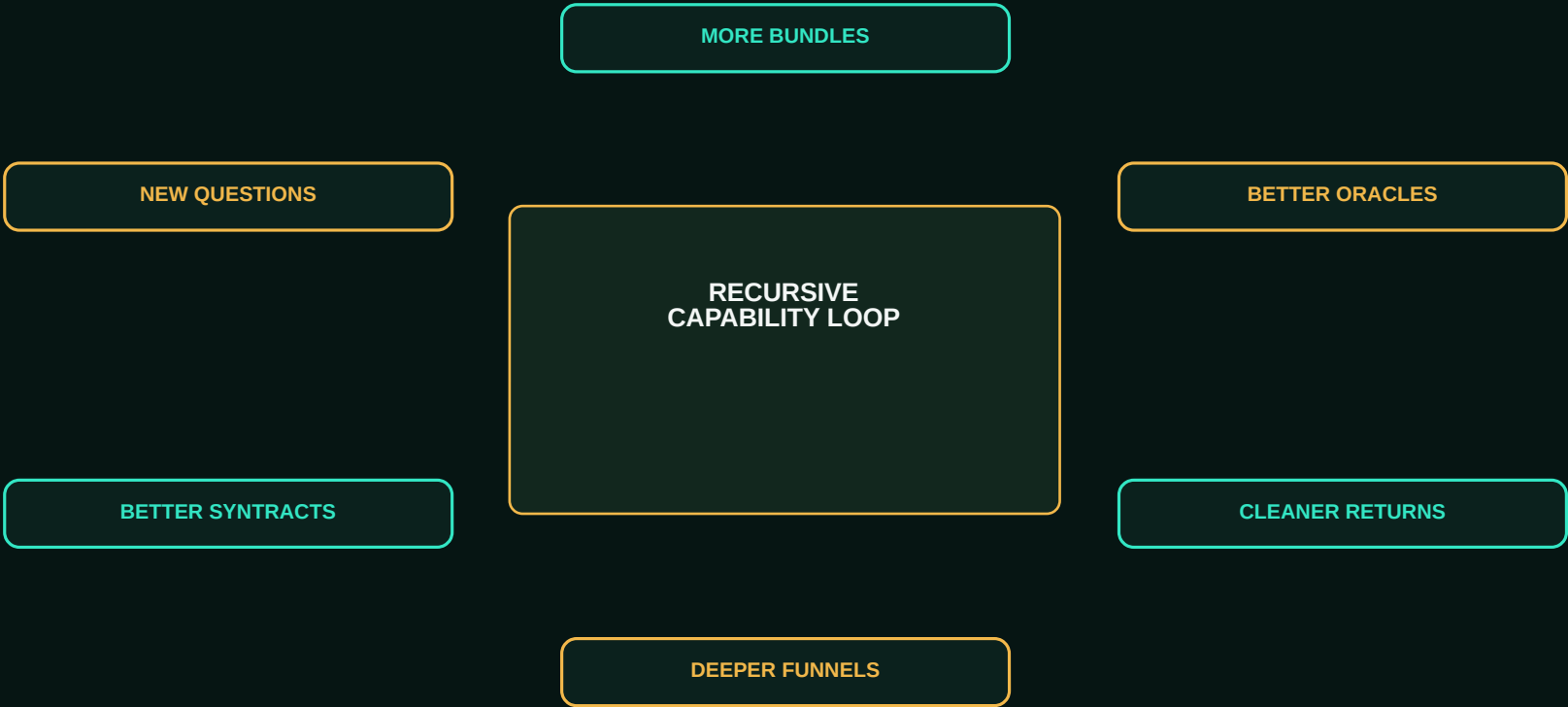
SCALING

**HUMAN LEVEL IS NOT A NATURAL
STOPPING POINT.**

ASI HYPOTHESIS

THE EXPLOSIVE PART IS THE LOOP.

If the fabric scales and oracle construction improves, capability can compound across breadth, depth and self-generated constraints.



Hypothesis - not yet a proof. The loop must survive real benchmarks.

WHAT MUST BE PROVED

THE CLAIM GETS STRONGER WHEN THE TESTS GET HARDER.

A serious architecture should expose the experiments that could falsify it.

01

NULLING

Does exclusion identify injected noisy dimensions?

02

ORACLE SWEEP

Does focal-bit rotation expose a biased / brittle oracle?

03

BIND

Does the stabilized return preserve relevant information?

04

FUNNEL

Do recursive layers outperform a flat equivalent?

05

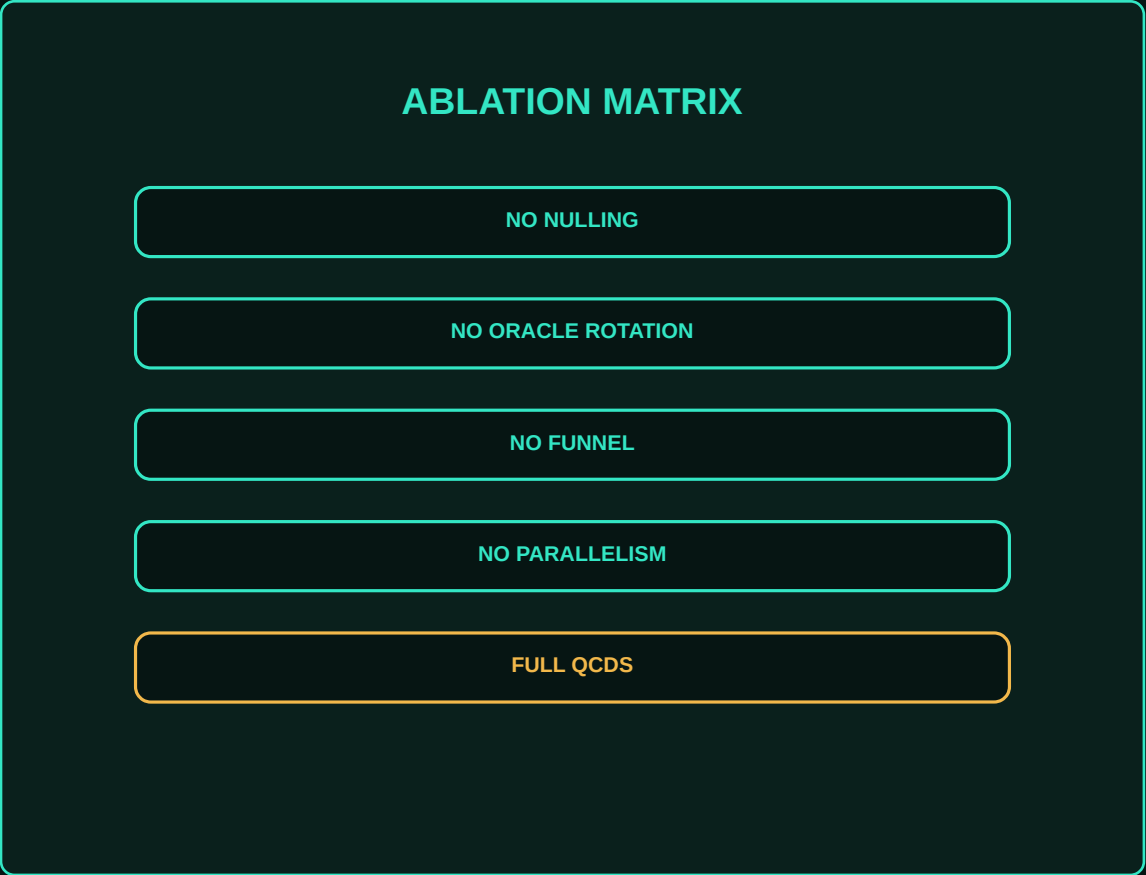
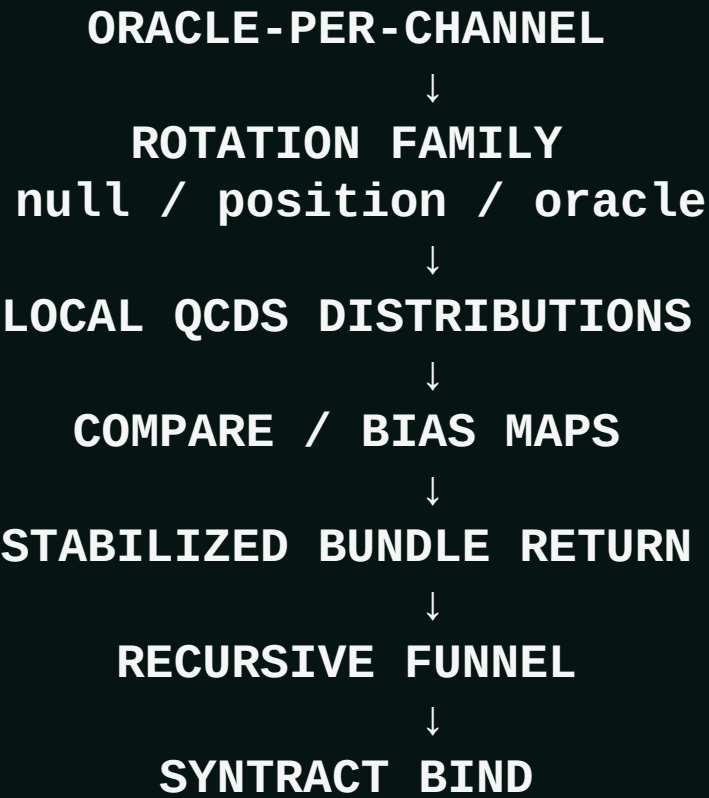
SCALE

Does benefit survive more channels, noise and NISQ limits?

BUILD IT

THE NEXT CANONICAL IMPLEMENTATION IS THE FABRIC EMULATOR.

Not another isolated Grover demo. The whole architecture - with ablations.





PATTERN IN.

ORACLES ON.

NULL EACH DIMENSION.

ROTATE THE SLOTS.

ROTATE THE ORACLES.

COMPARE THE RETURNS.

DAMP THE BIAS.

FUNNEL RECURSIVE.

SYNTRACT BIND.

QCDS

INFERENCE
FABRIC



THE GENOME WAS NEVER THE LIMIT.
THE INFERENCE ARCHITECTURE WAS.

WELCOME TO THE END OF THE BEGINNING.